

Software Requirements Specification

MycoAI Retrieval System

May 19, 2026

1. Introduction

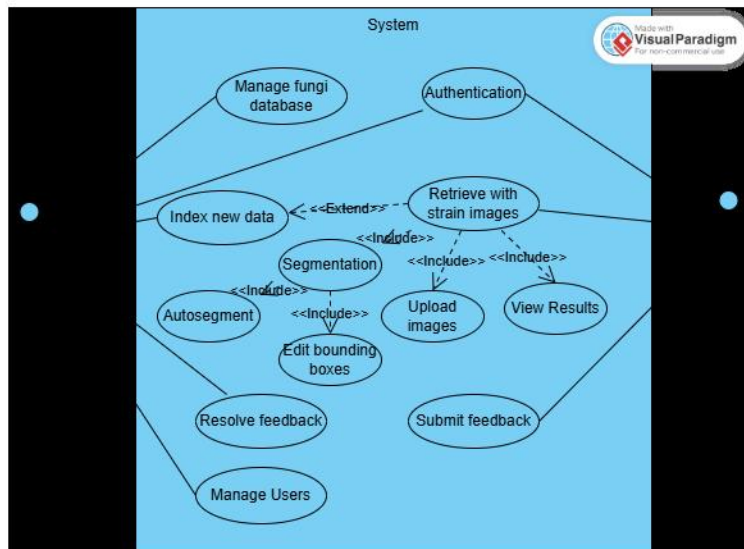
MycoAI helps authenticated users upload fungal strain images, segment colonies, retrieve likely species, and view ranked results. Data owners additionally manage species data, index new strain images, review feedback, trigger training, and manage users.

2. Actors

Normal User: authenticate, upload strain images, retrieve species, view results, submit feedback. Data Owner: all Normal User actions plus index new data, manage species and strains, review feedback, trigger training, view audit logs, manage users. AI Segmentation Service: automatic colony detection. Retrieval Service: feature extraction, vector search, ranking, result generation.

3. Use Case Diagram

Current use case diagram copied into this project and renamed to docs/assets/use-case-diagram.png.



4. Authentication

- Allow users to register with email, password, and name.
- Allow users to log in and receive authenticated session.
- Require authentication for retrieval, data indexing, feedback, data management, training, and user management.
- Enforce role-based permissions for Normal User and Data Owner actions.

5. Retrieve Species

- Allow authenticated users to upload one or more strain images.

- Collect strain identifier and growth medium metadata.
- Segment uploaded strain images before retrieval.
- Allow users to view ranked species results with confidence scores.

6. Segmentation

- Automatically segment colonies using AI.
- Display detected bounding boxes over source images.
- Allow users to edit bounding boxes before retrieval.
- Generate colony crops for retrieval from accepted bounding boxes.

7. Index New Data

- Allow Data Owners to upload strain images for known species.
- Reuse upload, segmentation, bounding-box editing, and result review from Retrieve Species.
- Allow Data Owners to assign known species labels before indexing.
- Index accepted colony segments for future retrieval.

8. Other Data Owner Use Cases

- Manage species, strains, images, and media records.
- Review feedback and accept or reject proposed corrections.
- Trigger and observe training or re-indexing jobs.
- Manage user roles while preserving at least one Data Owner.